A. **Floor and Convolutions.** The floor function offers rewrites of convolutions.

Theorem. Let f be arithmetic and $F =$

Let n be a positive integer. Then

$$\sum_{k=1}^{n} F(k) =$$

Proof. We examine the sum on the lefthand side.

$$\sum_{k=1}^{n} F(k) =$$

i.e. we are summing $f(d)$ for all

Returning to the general case, let's fix a specific value of d from 1 to n.

The number of times $f(d)$ will occur in our sum is

or in other words equals:

Thus, summing over d first, the sum is equivalent to

$$\sum_{d=1}^{n}$$

□

Example 1. Let n be an integer. Justify the following equalities.

$$\sum_{k=1}^{n} \tau(k) = \sum_{k=1}^{n} \lfloor n/k \rfloor$$

$$\sum_{k=1}^{n} \sigma(k) = \sum_{k=1}^{n} k \cdot \lfloor n/k \rfloor$$

Solution. Before we begin, we exhibit the τ equality in the case $n = 4$.

We recall:

Then we apply the theorem $\sum_{k=1}^{n} (f * \mathbb{1})(k) = \sum_{d=1}^{n} f(d) \left\lfloor \frac{n}{d} \right\rfloor$ from the last page.

B. **Euler Phi Function.** We count the number of relatively prime residues.

The **Euler phi** function is defined on positive integers n by:

$\phi(n) =$

For example, if $n = 12$, then

$\phi(12) =$

Or, if p is prime, then

$\phi(p) =$

and if p^k is a pos. power of a prime

$\phi(p^k) =$

in particular

$\phi(2^k) =$

Theorem. ϕ is multiplicative.

Proof. Evidently $\phi(1) =$

Now let $m \perp n$ be positive integers.

By the Chinese remainder theorem, each a in $[1 \dots mn]$ is uniquely determined

This establishes a one–to–one correspondence

Under this correspondence, a shares a common prime factor with mn iff

and thus:

Therefore: $[\#\text{ of } a\text{'s} \perp mn] =$

In other words:

□

Example 2. Let n be an integer greater than 1 with prime factorization

$n = p_1^{k_1} \cdots p_r^{k_r}$

Provide a formula for $\phi(n)$.

Solution. We use $\phi(p^k) = p^k - p^{k-1}$ and multiplicativity to obtain

Example 3. Find $\phi(360)$.

Solution. We compute